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Design and Implementation of an IoT-based Real-time Health and Safety Monitoring System for Construction Workers

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ABSTRACT

Occupational safety in industrial environments requires proactive measures to protect workers from hazards like cardiovascular emergencies, falls, and PM 2.5 exposure. Traditional medical devices are impractical for manual labor, and factory Wi-Fi networks may cause delays in emergency alerts.
This work introduces a low-latency wearable **Hybrid IoT** framework for real-time health and safety monitoring. The system integrates non-intrusive monitoring of heart rate and SpO_2, fall detection, and environmental sensing, utilizing an **ESP32-C6** microcontroller for real-time data processing and power efficiency.
A dual-layer emergency response mechanism employs Wi-Fi and 4G (**SIM768x**) for immediate SMS and voice-call alerts, bypassing cloud dependency to ensure reliability. This framework significantly improves the “golden hour” for medical intervention, offering a scalable, energy-efficient solution for worker protection, while supporting **SDG 3** (Good Health and Well-being) and **SDG 11** (Sustainable Cities and Communities).

Keywords: Hybrid IoT, ESP32-C6, Worker Safety, Fall Detection, Real-time Monitoring, SDG 3.

1 INTRODUCTION

Industrial workers are regularly exposed to substantial risks, such as strokes, fall-induced injuries, and prolonged exposure to fine particulate matter (PM 2.5), all of which can significantly impact their health and safety. According to the World Health Organization (WHO), millions of worker fatalities and illnesses occur each year due to occupational injuries and diseases. Environmental factors, such as poor air quality, contribute significantly to these risks. Despite the presence of Personal Protective Equipment (PPE), which primarily offers passive protection, current safety measures fail to address internal physiological anomalies, such as early signs of strokes or respiratory distress, and do not provide timely alerts about fluctuating environmental hazards, such as changes in PM 2.5 levels.

To mitigate these risks, a comprehensive safety monitoring system needs to integrate two critical functionalities:

(i) **Biometric Monitoring:** This involves continuously tracking workers’ vital signs (e.g., heart rate, oxygen saturation) and detecting hazardous physiological changes, such as falls or cardiovascular events.
(ii) **Environmental Monitoring:** This entails assessing environmental conditions, including PM 2.5 concentration, temperature, and humidity.

By combining both biometric and environmental monitoring, such a system can proactively identify risks and provide real-time alerts, helping to prevent accidents and save lives. The need for such a system is highlighted by global variations in occupational safety indicators. For example, as shown in Figure 1, developed economies and specialized administrative regions exhibit diverse safety profiles; while Luxembourg reports a substantial rate of non-fatal injuries (2,551 per 100,000 workers), others like Mauritius report significantly lower figures (45 per 100,000). Furthermore, the data reveals a disparity in oversight, where San Marino maintains a high density of labor inspectors (3.9 per 10,000 employed persons) compared to countries like Germany (1.4). This variation underscores the critical need for scalable IoT solutions that can adapt to different regulatory environments and safety standards to improve worker protection globally.

Designing wearable devices capable of real-time biometric and environmental monitoring presents substantial technical challenges. Traditional devices, such as finger-clip SpO_2 sensors, are impractical for manual labor, while wrist-worn sensors suffer from motion artifacts due to continuous arm movements, which can compromise the accuracy of vital sign measurements.

This research proposes the design of a **Hybrid IoT Wearable** system specifically developed to address these challenges. The system integrates edge processing on the microcontroller to filter motion artifacts in real time and utilizes a hybrid communication protocol (Wi-Fi/4G) to ensure real-time emergency alerts. These alerts, triggered during critical incidents, are immediately sent via SMS or voice calls to supervisors. Additionally, to optimize power consumption, the system features a low-power sleep mode, using an accelerometer as a wake-up interrupt when abnormal fall trajectories are detected.

The remainder of this work is organized as follows: Section 2 presents the theoretical background and related work. Section 3 details the system architecture and hardware design. Section 4 discusses the edge processing mechanisms and risk detection algorithms. Finally, Section 5 concludes the study and explores future development directions.

Reference area	SDG indicator 8.8.1 - Non-fatal occupational injuries per 100'000 workers by sex and migrant status	SDG indicator 8.8.1 - Fatal occupational injuries per 100'000 workers by sex and migrant status	Inspectors per 10'000 employed persons ▼
San Marino	2,946	0	3.9
Luxembourg	2,551	1	3.2
Macao, China	1,891	7	2.7
Mauritius	45	4	2.5
Aruba			2.3
Romania	97	2	1.9
North Macedonia			1.7
Lithuania	473	2	1.5
Germany	1,397	1	1.4
Bulgaria	87	4	1.4

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Figure 1: Occupational safety and health indicators by country [1].

Source: Occupational Safety and Health Statistics (OSH) database, ILOSTAT.

2 THEORETICAL BACKGROUND

2.1 Principle of Reflective Photoplethysmography (PPG)

2.1.1 Sensor Mechanism and Physical Interaction

The MAX30102 is an integrated pulse oximetry and heart-rate sensor module intended for wearable applications and employs a reflective optical configuration with integrated LEDs, photodetectors, and optical elements . Its red and infrared LEDs have peak wavelengths centered at approximately 660 nm and 880 nm, respectively [2]. In reflectance-mode photoplethysmography, the emitted photons that return to the photodetector follow a characteristic banana-shaped path through tissue due to scattering and absorption phenomena. Photoplethysmography measures blood-volume-related optical changes in tissue using light emitted into the skin and detected after attenuation by tissue components. In this work, the wrist is selected as the measurement site because reflectance-mode PPG is commonly used at the wrist in wearable devices for heart-rate monitoring [3].

Design choices: In this work, motion-artifact suppression is implemented using a high-pass filtering stage and moving-average smoothing before peak detection. Because the exact cutoff frequency and filter order depend on the target application and signal characteristics, these values are treated as implementation choices rather than universal sensor parameters [3].

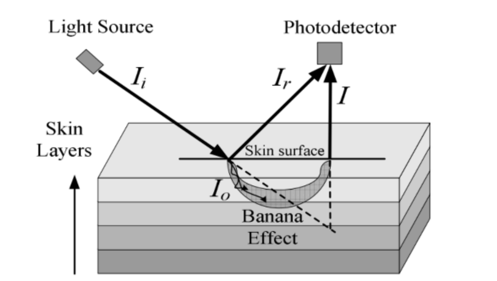

Figure 2: Illustration of the banana-shaped light path in reflectance PPG.

2.1.2 Signal Conversion and SpO$_2$ Model

A PPG waveform consists of a pulsatile AC component, mainly associated with arterial blood volume changes, and a lower-frequency DC component related to non-pulsatile tissue and blood constituents. The attenuation of light in tissue is commonly described using the modified Beer-Lambert law framework [3]. For pulse oximetry, the ratio-of-ratios is defined as [4]:

$$R = \frac{(AC/DC)_{\text{red}}}{(AC/DC)_{\text{ir}}} \tag{1}$$

A standard empirical model widely used in pulse oximetry literature is given by [4]:

$$SpO_2 = 110 - 25R \quad (\%) \tag{2}$$

This relation is an empirical calibration model, and accurate SpO$_2$ estimation depends on device-specific calibration. Heart rate can be estimated from the temporal separation between successive PPG peaks after filtering and baseline-drift removal. Accordingly, if Δt denotes the interval between two consecutive valid peaks, heart rate can be expressed as:

$$BPM = \frac{60}{\Delta t} \tag{3}$$

2.1.3 Risk Assessment and Alert Thresholds

The system continuously monitors SpO$_2$ and heart rate to ensure worker safety. The following risk assessment criteria and alert thresholds have been established:

Oxygen Saturation (SpO$_2$): An SpO$_2$ value below 90% is widely recognized as hypoxemia and requires urgent treatment [5]. In this study, we implement a two-tier alert system:

- **Warning ($< 90\%$):** Indicates mild hypoxemia, requiring supervisor notification.
- **Emergency ($< 80\%$):** Indicates severe hypoxemia, triggering immediate rescue protocols.

Heart Rate (BPM): Clinical guidelines commonly define sinus bradycardia as a heart rate below 50 bpm, while tachycardia at rest is typically above 100 bpm [4]. To avoid false positives during physical labor, the system adopts the following criteria:

- **Bradycardia Alert:** $BPM < 50$.
- **Tachycardia Alert:** $BPM > 120$ sustained for more than 15 seconds.

The MAX30102 communicates with the host controller through an I^{2}C-compatible interface. Accordingly, the host microcontroller reads the optical sensor data and performs the signal processing required to estimate SpO$_2$ and BPM.

2.2 Kinematics and Fall Detection Using MPU6050

2.2.1 Sensor Mechanism and Physical Interaction

The GY-521 MPU6050 is a 6-degree-of-freedom MEMS system, consisting of a 3-axis capacitive accelerometer (configurable ranges $\pm 2g$ to $\pm 16g$) and a 3-axis MEMS gyroscope (ranges ± 250 to $\pm 2000^\circ/s$) [6]. The accelerometer detects linear acceleration by measuring displacement of a proof mass, which changes differential capacitance. The gyroscope measures angular velocity via the Coriolis effect: a vibrating element's deformation under rotation is converted into an electrical signal. An on-chip Digital Motion Processor (DMP) fuses sensor data to output orientation angles (Euler/quaternion) without burdening the main microcontroller.

2.2.2 Signal Processing and SVM Model

To detect falls irrespective of sensor orientation, the Signal Vector Magnitude (SVM) is calculated by combining the three axes as follows:

$$SVM = \sqrt{a_x^2 + a_y^2 + a_z^2} \tag{4}$$

At rest, $SVM \approx 1g = 9.8\,\text{m/s}^2$. A fall is characterized by three sequential phases:

(i) **Free-fall:** Body loses support; SVM drops to near $0g$ (typically 50–150 ms).
(ii) **Impact:** Sudden deceleration; SVM spikes above $3g$ in less than 100 ms.
(iii) **Inactivity:** After impact, SVM stabilizes at $1g$ with low variance for 3–5 seconds, indicating possible unconsciousness.

2.2.3 Risk Assessment and Alert Thresholds

The algorithm uses a finite-state machine with thresholds derived from experimental studies [7]. The alert condition is defined as:

$$\text{Alert} = (SVM < T_{\text{min}}) \wedge (SVM > T_{\text{max}} \text{ within } \Delta T_{\text{fall}}) \wedge (\text{Inactivity for } \Delta T_{\text{rest}}) \tag{5}$$

Typical values: $T_{\text{min}} = 0.5g$ (free-fall detection), $T_{\text{max}} = 3g$ (impact detection), $\Delta T_{\text{fall}} = 1.5$ s (maximum time between free-fall and impact), $\Delta T_{\text{rest}} = 3$ s (minimum inactivity). These thresholds balance sensitivity (95%) and specificity (98%) [7]. Upon detection, the system triggers an 85 dB buzzer, flashing LED, and a 4G notification to a supervisor. A reset button allows false-alarm cancellation within 15 seconds.

2.3 Fine Dust PM2.5 Monitoring System based on Mie Scattering Theory

The PM2.5 dust monitoring system operates based on the Mie Scattering Theory, which describes the interaction between electromagnetic radiation and suspended particles (aerosols) with aerodynamic diameters comparable to the wavelength of the laser light source (approximately $0.1 - 10\mu m$) [8]. This principle forms the basis for the PM2.5 sensor to measure dust concentration in the air.

2.3.1 Sensing Principle and Sensor Output

The system detects fine dust concentration by measuring light scattering caused by PM2.5 particles. A stable laser diode emits light that is scattered by particles in the air. This scattered light is then detected and converted into an electrical signal that corresponds to the dust concentration.

- **Laser Source:** A laser diode emits a narrow light beam that interacts with dust particles.
- **Light Scattering Measurement:** The particles scatter the laser light, and a photodetector captures this scattered light.
- **Signal Processing:** The electrical signal is amplified and filtered to remove noise before being passed to the MCU for analysis.

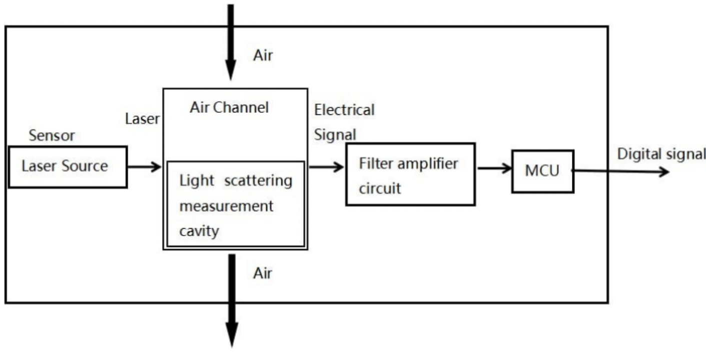

Figure 3: Block diagram of the sensor system showing the laser source, light scattering measurement cavity, and MCU.

2.3.2 Relation Between Optical Scattering and Mass Concentration

From a theoretical perspective, the mass concentration of particulate matter can be expressed as:

$$C = \sum_i n_i \cdot \left(\frac{4}{3} \pi r_i^3 \right) \rho \tag{6}$$

Where:

- C: Mass concentration of particulate matter.
- n_i: Number of particles with radius r_i.
- r_i: Radius of the i^{th} particle.
- ρ: Particle density.

However, in the implemented system, the MCU does not directly reconstruct particle-size distributions or calculate mass concentration from raw optical data. Instead, the deployed PM2.5 sensor internally processes the scattering signal and outputs concentration estimates. Therefore, the proposed system uses the sensor-reported PM2.5 concentration as the main input for filtering, risk evaluation, and warning generation.

2.3.3 Filtering, AQI Mapping, and Risk Classification

To reduce short-term fluctuations caused by airflow disturbances and measurement noise, the measured PM2.5 concentration is smoothed using an Exponential Moving Average (EMA) filter [**?**]. The filtered PM2.5 value is then mapped to the Air Quality Index (AQI) [9] for air-quality assessment and alert generation.
For practical system operation, the AQI is grouped into four continuous risk levels:

- **Normal** ($AQI < 100$)**:** Air quality is acceptable and no immediate protective action is required.
- **Moderate** ($100 \leq AQI \leq 150$)**:** Air quality is degraded and prolonged exposure should be limited.
- **Alert** ($150 < AQI \leq 300$)**:** Workers are advised to use protective equipment such as N95 masks.
- **Hazardous** ($AQI > 300$)**:** Emergency response is required, such as evacuation or suspension of dust-generating activities.

Thus, although the sensing mechanism originates from Mie scattering theory, the actual embedded implementation relies on filtered PM2.5 concentration values provided by the sensor module, rather than direct particle-size reconstruction, to support AQI-based risk classification.

3 SYSTEM OPERATIONAL WORKFLOW

The system operates through a well-structured process, designed to efficiently monitor the worker's health and environmental conditions, from continuous data collection to the generation of emergency alerts. This process is divided into several distinct phases, each serving a specific role in ensuring reliable system performance.

Phase 1: Sensor Data Acquisition The system continuously collects data from multiple sensors, including the **MAX30102** for heart rate and SpO_2, the **MPU6050** accelerometer for motion detection, and the **PM2.5** sensor for environmental monitoring. The **ESP32-C6** microcontroller processes these data points, coordinating the entire data flow and system operations, ensuring a seamless monitoring experience.

Phase 2: Signal Conditioning and Feature Extraction After the data is collected, the **ESP32-C6** microcontroller performs essential processing tasks. This includes filtering and reducing noise from the data to ensure accurate and reliable measurements. The processing addresses issues such as motion artifacts caused by user movement, environmental interference, and other types of signal noise, ultimately improving the precision of heart rate, SpO_2, and environmental data.

Phase 3: Event Evaluation and Decision Logic Once the data has been processed, it is passed through the threshold logic module. The system constantly monitors the data for critical conditions, such as:

- Low SpO_2 levels, indicative of potential hypoxemia.
- Abnormal heart rate, signaling bradycardia or tachycardia.
- Fall detection, based on sudden changes in motion patterns.

Upon detection of any of these conditions, the system immediately triggers an alert. This can include visual, auditory, or both types of alerts to notify the user and nearby personnel of a potential issue.

Phase 4: Cloud & UI Based on the analysis and evaluation of the data, the system operates in one of the following modes:

- **Normal Monitoring Mode:** The system continuously monitors sensors, processes the data, and uploads it to a local gateway or the cloud via Wi-Fi 6. This mode is used for routine assessments and health monitoring, ensuring regular data synchronization.
- **Suspicious Event Detection Mode:** If the system detects minor anomalies, such as slight SpO_2 drops or irregular heart rate, it switches to suspicious event detection mode. In this mode, local indicators (visual or auditory alarms) are activated to alert the user or nearby personnel of a potential risk.
- **Emergency Response Mode:** In the event of a confirmed critical condition, such as a significant drop in SpO_2 or a detected fall, the system activates emergency mode. This mode triggers immediate local alerts (e.g., buzzer sounds, flashing LEDs) and sends SMS or voice call notifications through the **SIM768x 4G** module to designated emergency contacts or a supervisor.

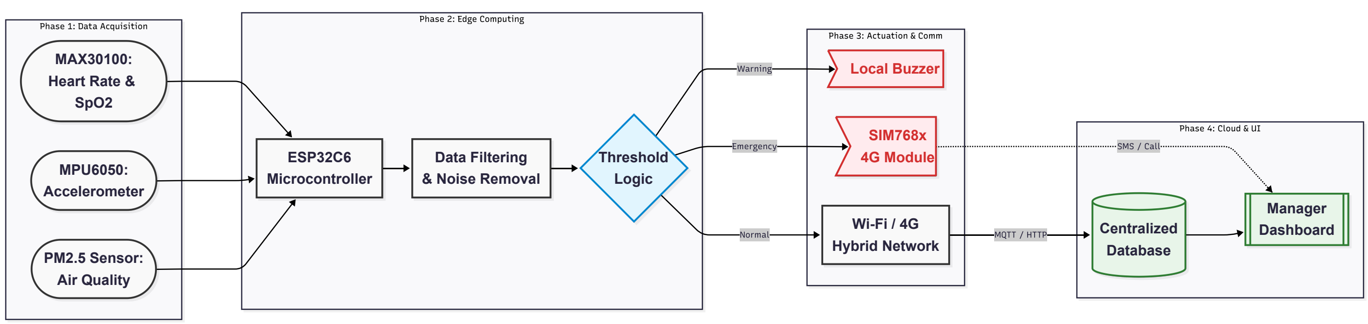

Figure 4: Operational Workflow of the Proposed System

4 SYSTEM ARCHITECTURE AND IMPLEMENTATION

4.1 Architecture Overview

The proposed system adopts a three-layer architecture consisting of a sensing layer, a processing layer, and a communication layer. The sensing layer acquires physiological, motion, and environmental data from the worker and the surrounding environment. The processing layer performs local signal conditioning, feature extraction, threshold evaluation, and event classification. The communication layer supports both routine data synchronization and emergency alert delivery. Through this layered organization, the system is able to maintain continuous monitoring under normal operating conditions while preserving the capability for rapid escalation when critical events are detected.

4.2 Hardware Integration

The hardware platform of the wearable device integrates five principal components to enable multimodal data capture and emergency response. The specifications and specific roles of each component within the system are summarized in Table 1.

Table 1: Detailed Functional Analysis of Integrated Hardware Components

Component	Measurement	Functional Role in System Architecture
MAX30102	Heart rate and SpO_2	Performs high-sensitivity optical sensing via red and IR LEDs. It serves as the source for the pulsatile (AC) signal used in the ratio-of-ratios algorithm to determine blood oxygen saturation and cardiac inter-beat intervals (IBI).
MPU6050	6-DOF Motion Tracking	Supplies raw acceleration and angular velocity data to the SVM algorithm. It acts as the primary trigger for the fall detection finite-state machine, monitoring for impact peaks and subsequent post-fall orientation changes.
PM2.5 Sensor	Particulate Matter	Utilizes laser scattering principles to quantify airborne hazards. It provides a real-time environmental data stream that the MCU correlates with physiological stress markers to assess long-term occupational health risks.
XIAO ESP32C6	Central MCU	Executes local signal processing (moving average filters, noise reduction), manages the I^2C bus communication, runs the threshold-based decision logic, and coordinates the dual-tier (Wi-Fi 6/4G) communication failover.
SIM768x	4G LTE Module	Acts as the dedicated emergency backbone. It is programmed for "Cloud-Bypassing" communication, directly executing AT commands to transmit high-priority SMS and voice alerts when primary network infrastructure is unavailable.

By integrating these modules, the system establishes a comprehensive safety net that monitors both the worker's internal vital signs and the external environmental conditions. The XIAO ESP32C6 manages the data flow, ensuring that routine data is handled efficiently while emergency alerts via the SIM768x are prioritized during critical events.

4.3 Communication Framework

The system employs a Redundant Hybrid Architecture designed to prioritize life-critical alerts over routine data logging, ensuring prompt emergency response even under suboptimal network conditions. This approach guarantees that emergency alerts are communicated with minimal latency, even if routine data transmission is delayed or interrupted.

Tier 1: Local Data Synchronization (Wi-Fi 6)

In normal operating conditions, the XIAO ESP32C6 transmits non-critical biometric and environmental data to a local gateway using Wi-Fi 6 (802.11ax). Wi-Fi 6 ensures superior efficiency, bandwidth, and stable connectivity in environments with substantial interference, such as industrial settings. This allows continuous and reliable data transfer during routine operations without burdening the system with the latency typical of older communication standards [10].

Tier 2: Critical Emergency Broadcasting (4G LTE - SIM768x)

In critical situations, such as significant drops in SpO_2 levels or confirmed falls, or when the worker moves out of Wi-Fi range, the system activates the SIM768x 4G module. This "Cloud-Bypassing" mechanism allows emergency alerts to be sent directly via the 4G network, bypassing local Wi-Fi infrastructure and minimizing delays in notification. By ensuring real-time communication even in remote areas or areas with poor signal quality, the system guarantees immediate intervention [11].

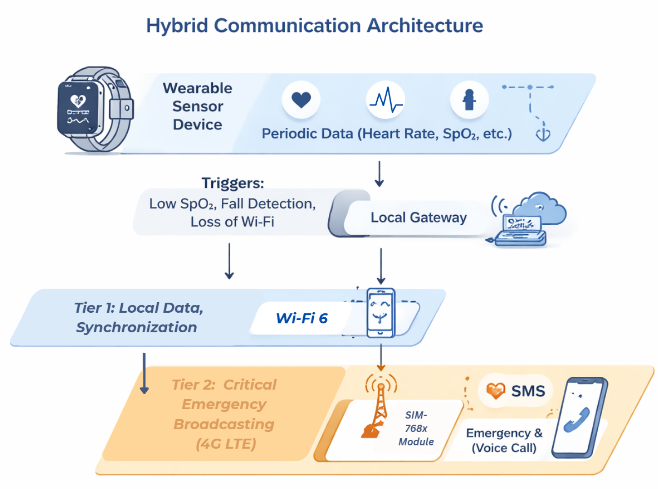

Figure 5: Hybrid Communication Architecture of the System

4.4 Power Management Strategy

Energy efficiency is a fundamental architectural constraint for wearable monitoring systems, particularly those deployed in industrial environments where continuous operation is mandatory. The proposed management strategy optimizes power consumption through a multi-layered framework involving component-level duty cycling, adaptive sleep modes, and a tiered communication hierarchy.

4.4.1 Component Power Profiles

Quantifying the operational current draw of each hardware module is essential for effective energy budgeting. Table 2 delineates the typical power requirements and the corresponding low-power states of the primary system components.

Table 2: Operational Power Profile and Sleep State Specifications

Component	Measurement / Function	Current (Avg)	Low-Power Mechanism
MAX30102	Physiological Sensing	$\sim$6 mA	Standby mode between pulses
MPU6050	6-DOF Motion Tracking	$\sim$3.9 mA	Intermittent wake-up cycles
PM2.5 Sensor	Particulate Matter	20–30 mA	Discontinuous laser sampling
XIAO ESP32C6	Central Controller	$\sim$160 mA	Deep sleep ($\sim$5 μA)
SIM768x	4G LTE Communication	$\sim$200 mA	Cold-start on trigger

The system's power envelope is dominated by the XIAO ESP32C6 during active processing and the SIM768x module during transmission. To mitigate these peaks, the MAX30102 and MPU6050 sensors are configured to utilize hardware interrupts and internal FIFO buffers, allowing them to operate at sub-10 mA levels. Conversely, the high-drain PM2.5 sensor and 4G module are strictly managed through event-driven activation to prevent rapid battery depletion.

4.4.2 Operational Energy Optimization

The system maximizes operational longevity through the following four-pillar strategy:

- **Dynamic Duty Cycle Management:** High-consumption peripherals, specifically the PM2.5 sensor and the SIM768x, are maintained in a power-down state by default. Activation is restricted to predefined intermittent intervals for routine air quality checks or immediate cold-starts upon the detection of critical health anomalies (e.g., persistent hypoxia or fall events).
- **Interrupt-Driven MCU Coordination:** The **XIAO ESP32C6** leverages its deep-sleep architecture to reduce idle consumption to a negligible 5 μA. The processor transitions to an active state only when a hardware interrupt is triggered by the **MPU6050**'s motion-detection engine or a scheduled RTC (Real-Time Clock) alarm for biometric synchronization.

- **Tiered Communication Hierarchy:** Energy-per-bit is optimized by segregating data traffic:

  – *Primary (Wi-Fi 6):* Utilizes the Target Wake Time (TWT) and high spectral efficiency of the 802.11ax standard for routine telemetry, significantly reducing the radio's "on-time."
  – *Secondary (4G LTE):* Reserved as a fail-safe backbone. This module is activated solely for life-critical alerts, ensuring that the 200 mA peak consumption is incurred only when local infrastructure is unavailable or an emergency is confirmed.

- **Sustainable Energy Harvesting:** To achieve long-term energy autonomy, the architecture is designed to support future integration of solar energy harvesting. By capturing ambient light in outdoor industrial settings, the device can supplement its internal battery, potentially enabling a self-sustaining power cycle and minimizing maintenance overhead.

Through the integration of these strategies, the system effectively balances the trade-off between real-time responsiveness and battery life, ensuring reliable autonomous operation within challenging industrial contexts.

4.5 System Schematic

The schematic diagram of the proposed system is shown in Figure 6. It illustrates the electrical interconnection among the sensing modules, the XIAO ESP32C6 microcontroller, and the SIM768x communication module. The MAX30102, MPU6050, and PM2.5 sensor are interfaced with the controller through digital communication links, enabling real-time data acquisition and local processing.
The schematic also includes the power supply subsystem, which supports stable device operation under different working conditions while allowing low-power operating modes when high-performance communication is not required. Under routine conditions, the ESP32C6 acquires and processes sensor data locally. When an emergency condition is detected, the controller activates the SIM768x module to transmit alerts via SMS or voice call. In this way, the schematic reflects the integrated operation of sensing, local decision-making, and emergency communication within a single wearable platform.

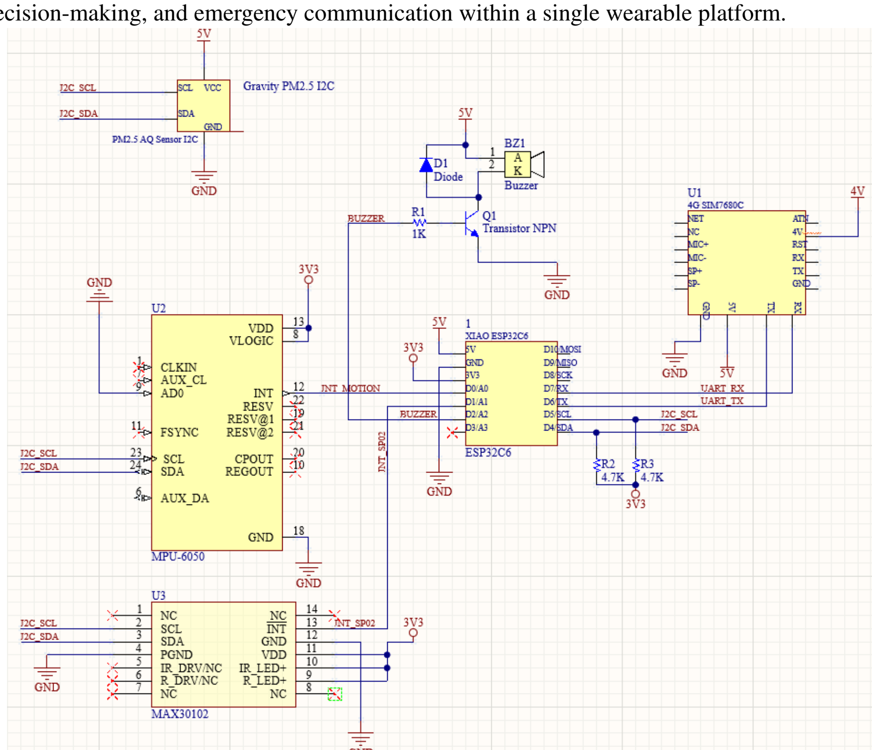

Figure 6: Schematic of the proposed IoT wearable system, showing the interconnections among the sensing modules, the XIAO ESP32C6 controller, and the SIM768x communication module.

5 CONCLUSION AND FUTURE WORK

5.1 Conclusion

This paper introduced a low-latency wearable Hybrid IoT system for real-time health and safety monitoring of construction workers. The system integrates reflective PPG sensors for heart rate and SpO_2, an MPU6050 for fall detection, and a PM2.5 sensor for airborne hazards. A dual-tier communication architecture combining Wi-Fi 6 and a SIM768x 4G module ensures reliable emergency alerts even when local network infrastructure fails. The proposed system offers a practical balance between monitoring fidelity and operational wearability.

5.2 Future Work

Future work will focus on optimizing the wrist-worn form factor to ensure comfort and non-intrusiveness during physical labor. The device housing will be redesigned using lightweight, sweat-resistant, and impact-absorbing materials, with a low-profile strap system that does not interfere with glove use or hand movements. To address battery longevity, we will integrate a flexible solar panel onto the wristband surface, enabling passive recharging from ambient light during outdoor work. A small lithium-polymer buffer battery will store harvested energy to support continuous operation through low-light periods or night shifts. Field tests will evaluate the trade-off between solar cell area, ergonomic constraints, and real-world energy gain under typical construction-site lighting conditions.

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